

Combined QA - R10C · D10 · L002 · 2021-09-15

Flightline: NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance | Overall status: WARN | Schema 1.3

Stage Summary

Source acquisition: PASS | valid within footprint=not available | median=not available | negative=not available | known bad bands=not available

Input reflectance: WARN | valid within footprint=1 | median=0.2504 | negative=0 | known bad bands=68

Correction parameters: WARN | valid within footprint=not available | median=not available | negative=not available | known bad bands=not available

BRDF and topographic correction: WARN | valid within footprint=1 | median=0.2504 | negative=0 | known bad bands=68

Spectral convolution: PASS | valid within footprint=1 | median=0.14445 | negative=0 | known bad bands=not available

Parquet extraction and merge: PASS | valid within footprint=not available | median=not available | negative=not available | known bad bands=not available

Report Artifacts

HTML: qa/combined/combined_qa.html

JSON: qa/combined/combined_qa.json

PDF: qa/combined/combined_qa.pdf

Cross-stage Interpretation

Flightline: NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance

What We Learn

WARN: The pipeline completed with stage-level warnings. Evidence: {"stages": ["Input reflectance", "Correction parameters", "BRDF and topographic correction"]}

WARN: Known poor-quality wavelength bands were retained and explicitly labeled; no masking or replacement was applied.

Evidence: {"stages": [{"classification_mode": "report_only_no_masking", "data_modified": false, "known_bad_band_count": 68, "stage": "Input reflectance"}, {"classification_mode": "report_only_no_masking", "data_modified": false, "known_bad_band_count": 68, "stage": "BRDF and topographic correction"}]}

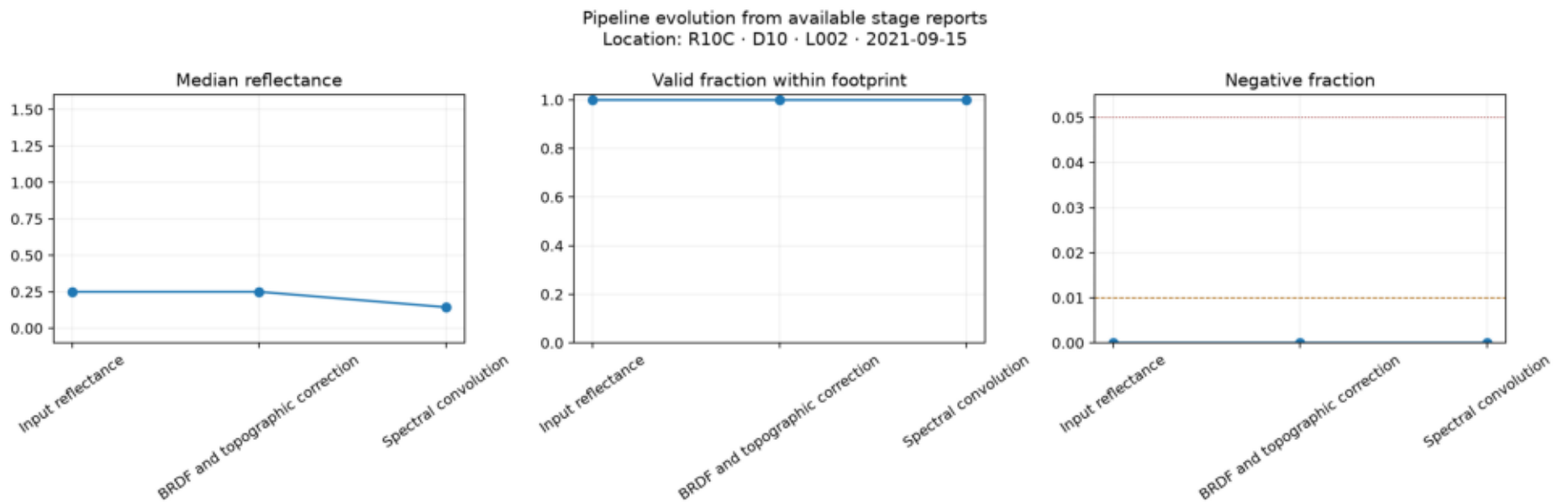
Not Evaluated

sensor_triangle_path_and_cycle_consistency: No fitted translation-edge artifacts are produced by the canonical NEON processing run.

landsat_direct_nbar_comparability: Direct Landsat Collection 2 NBAR is acquired outside this NEON flightline pipeline.

Combined QA Plot

pipeline_evolution.png



Source acquisition - PASS

Stage ID: acquisition | Mode: deep | Report: qa/stages/00_acquisition/stage_qa.html

Automated Interpretation

All evaluated checks passed their current thresholds.

Checks

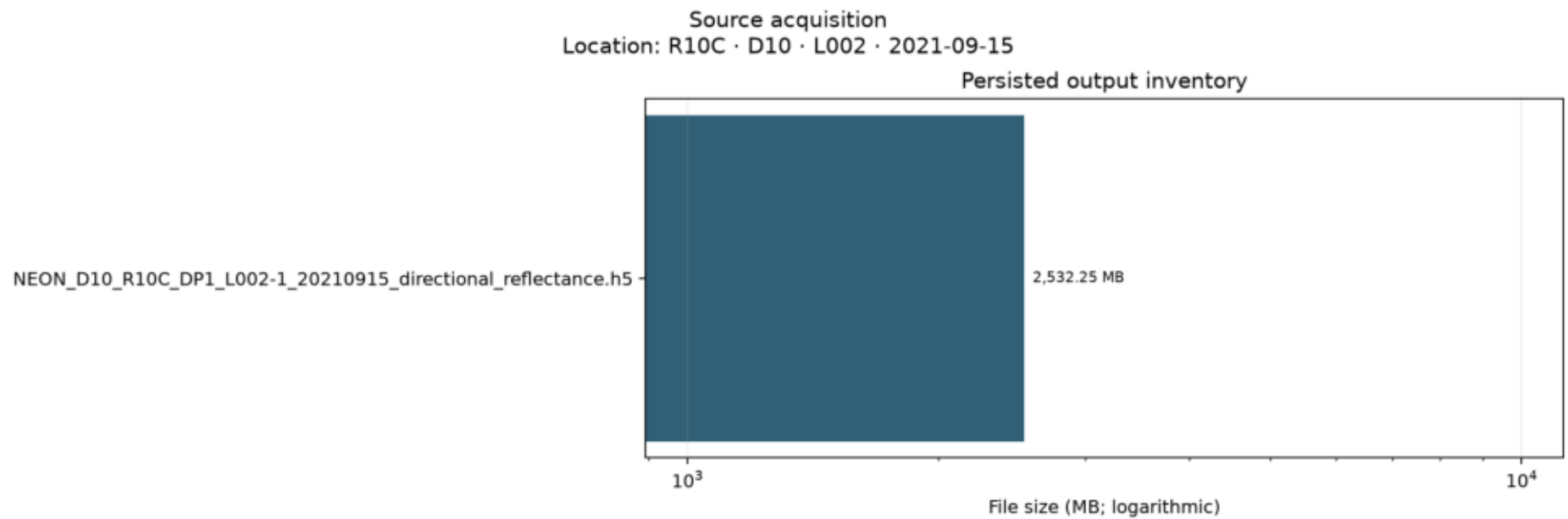
output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance.h5: PASS | value=True | The canonical stage output exists on disk. |

Unavailable Diagnostics

None.

Source acquisition Plot

Source acquisition: overview.png



Input reflectance - WARN

Stage ID: input_data | Mode: deep | Report: qa/stages/01_input_data/stage_qa.html

Automated Interpretation

Review provisional warnings: known_bad_spectral_bands_retained.

1 diagnostic(s) were explicitly not evaluated; see reasons below.

Checks

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

within_footprint_valid_reflectance_fraction: PASS | value=1 fraction | Valid spectral support should remain high inside the observed flight footprint; structural background outside it is reported separately. |

negative_reflectance_fraction: PASS | value=0 fraction | Large negative fractions may indicate correction or scaling artifacts. |

usable_band_reflectance_above_1_2_fraction: PASS | value=1.41814e-05 fraction | Unexpected values above 1.2 are evaluated on wavelengths not already labeled as known poor-quality regions; the all-band fraction remains reported. |

known_bad_spectral_bands_retained: WARN | value=68 bands | Established poor-quality wavelength regions are labeled in QA and retained in the product. | No masking, filtering, replacement, or file modification was applied.

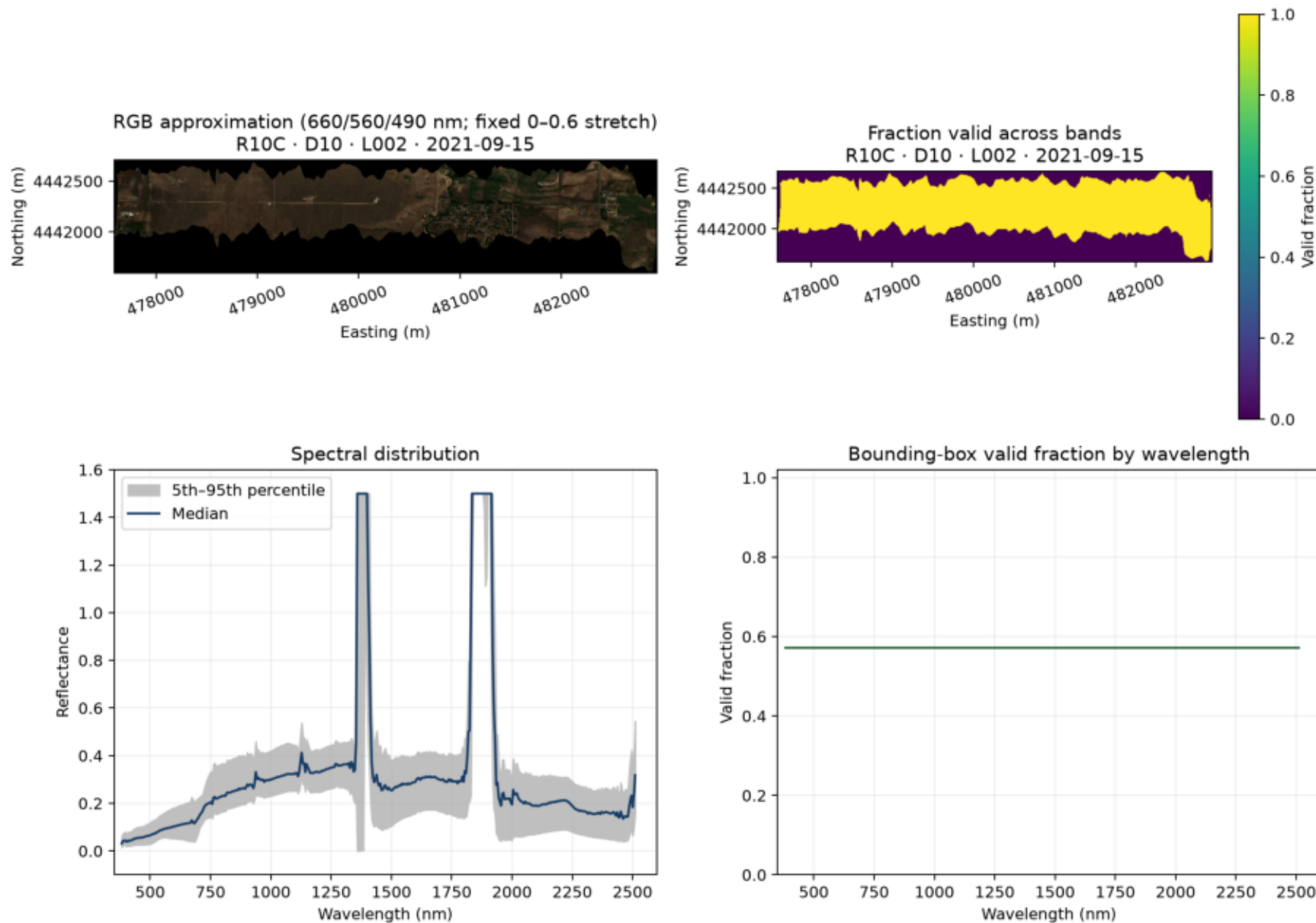
Unavailable Diagnostics

cloud_shadow_water_saturation_masks: The raw ENVI contract does not encode all source QA mask classes in a common field.

Input reflectance Plot

Input reflectance: overview.png

Input reflectance
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Correction parameters - WARN

Stage ID: correction_parameters | Mode: deep | Report: qa/stages/02_correction_parameters/stage_qa.html

Automated Interpretation

Review provisional warnings: persisted_geometry_physical_range_review.

Checks

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_brdfandtopo_corrected_envi.json: PASS | value=True |

The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_brdf_model.json: PASS | value=True | The canonical stage output exists on disk. |

geometry_field_fraction: PASS | value=1 fraction | Correction QA requires the physical geometry fields used by the model. |

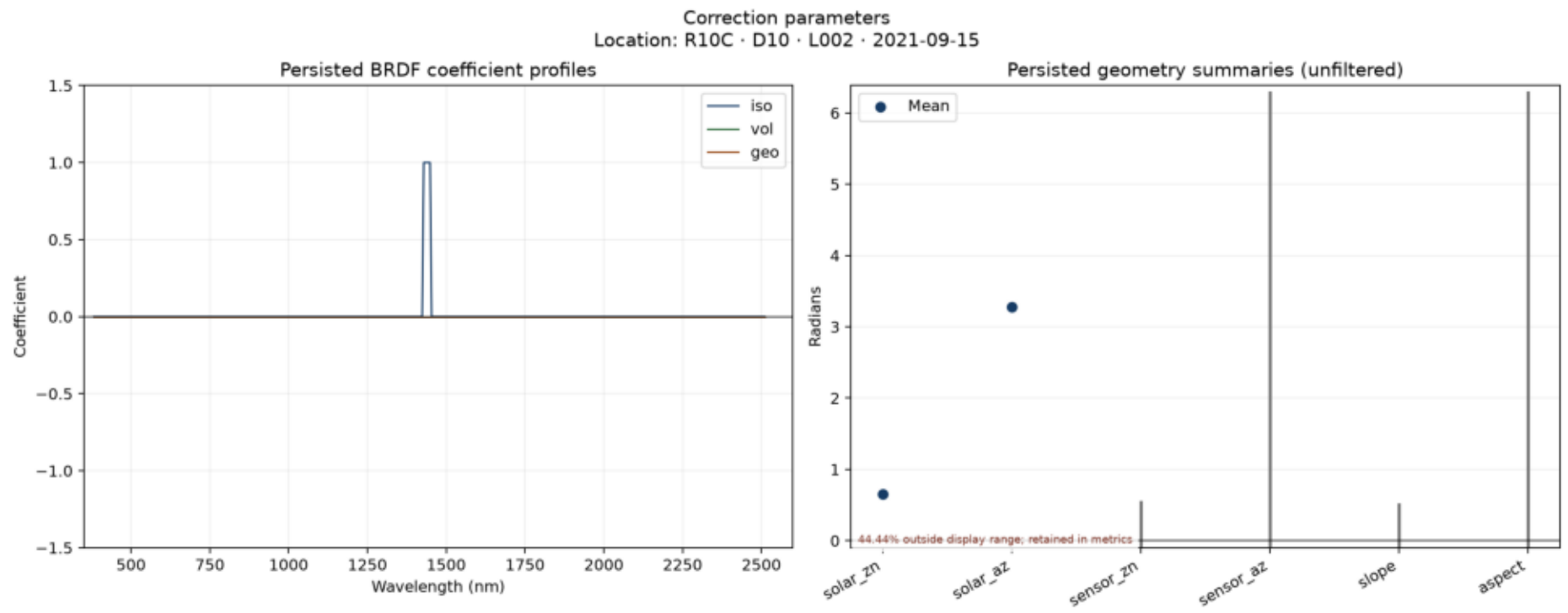
persisted_geometry_physical_range_review: WARN | value=4 fields | Persisted geometry min/mean/max values should lie within their physical radian ranges. | Out-of-range summaries are retained and shown; they may indicate source no-data contamination and are not masked by QA.

Unavailable Diagnostics

None.

Correction parameters Plot

Correction parameters: overview.png



BRDF and topographic correction - WARN

Stage ID: brdf_topographic_correction | Mode: deep | Report: qa/stages/03_brdf_topographic_correction/stage_qa.html

Automated Interpretation

Review provisional warnings: known_bad_spectral_bands_retained.

4 diagnostic(s) were explicitly not evaluated; see reasons below.

Checks

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_brdfandtopo_corrected_envi.img: PASS | value=True |

The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_brdfandtopo_corrected_envi.hdr: PASS | value=True |

The canonical stage output exists on disk. |

within_footprint_valid_reflectance_fraction: PASS | value=1 fraction | Valid spectral support should remain high inside the observed flight footprint; structural background outside it is reported separately. |

negative_reflectance_fraction: PASS | value=0 fraction | Large negative fractions may indicate correction or scaling artifacts. |

usable_band_reflectance_above_1_2_fraction: PASS | value=1.41415e-05 fraction | Unexpected values above 1.2 are evaluated on wavelengths not already labeled as known poor-quality regions; the all-band fraction remains reported. |

known_bad_spectral_bands_retained: WARN | value=68 bands | Established poor-quality wavelength regions are labeled in QA and retained in the product. | No masking, filtering, replacement, or file modification was applied.

absolute_correction_q99: PASS | value=0.00389395 reflectance | Extreme correction magnitude can signal overcorrection or invalid support. |

maximum_chunk_seam_score_after: NOT EVALUATED | value=not available ratio | Chunk-boundary gradients should resemble ordinary neighboring-pixel gradients. | This run has no internal application boundary or the chunk layout was unavailable.

Unavailable Diagnostics

chunk_seam_score: No internal application boundary was present or recorded for this stage.

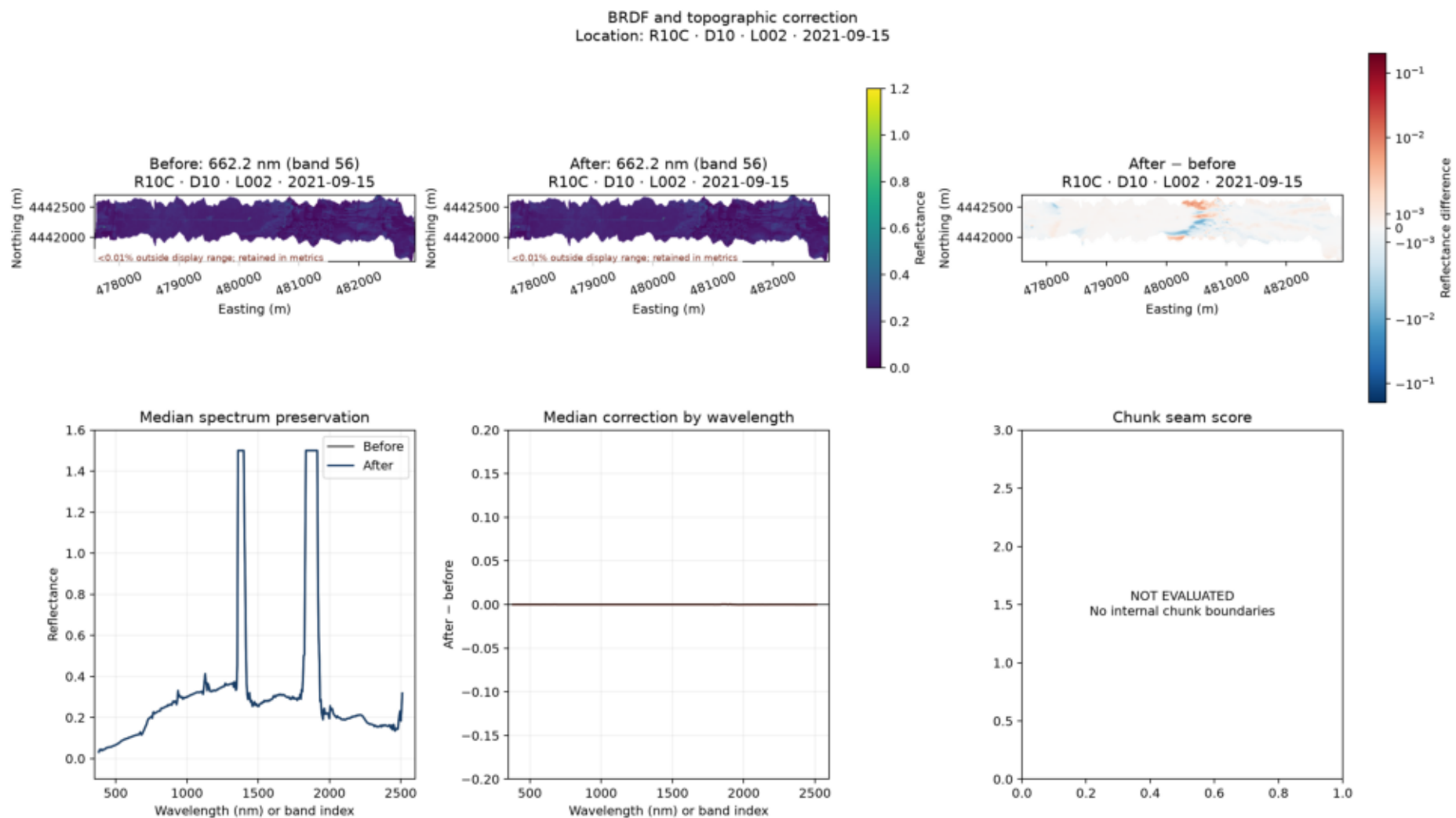
separate_topographic_and_brdf_attribution: The canonical stage persists only the combined corrected cube, not a topographic-only intermediate.

illumination_and_geometry_residual_models: Standard QA does not reopen the source HDF5 ancillary rasters; enable a future ancillary-aware deep run.

chunk_invariance_rerun: This report inspects the produced cube but does not rerun correction under an alternate chunk configuration.

BRDF and topographic correction Plot

BRDF and topographic correction: overview.png



Spectral convolution - PASS

Stage ID: spectral_convolution | Mode: deep | Report: qa/stages/04_spectral_convolution/stage_qa.html

Automated Interpretation

All evaluated checks passed their current thresholds.

2 diagnostic(s) were explicitly not evaluated; see reasons below.

Checks

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_etm+_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_etm+_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli2_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli2_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_tm_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_tm_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_oli_oli2_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_oli_oli2_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_tm_etm+_envi.img: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_tm_etm+_envi.hdr: PASS | value=True | The canonical stage output exists on disk. |

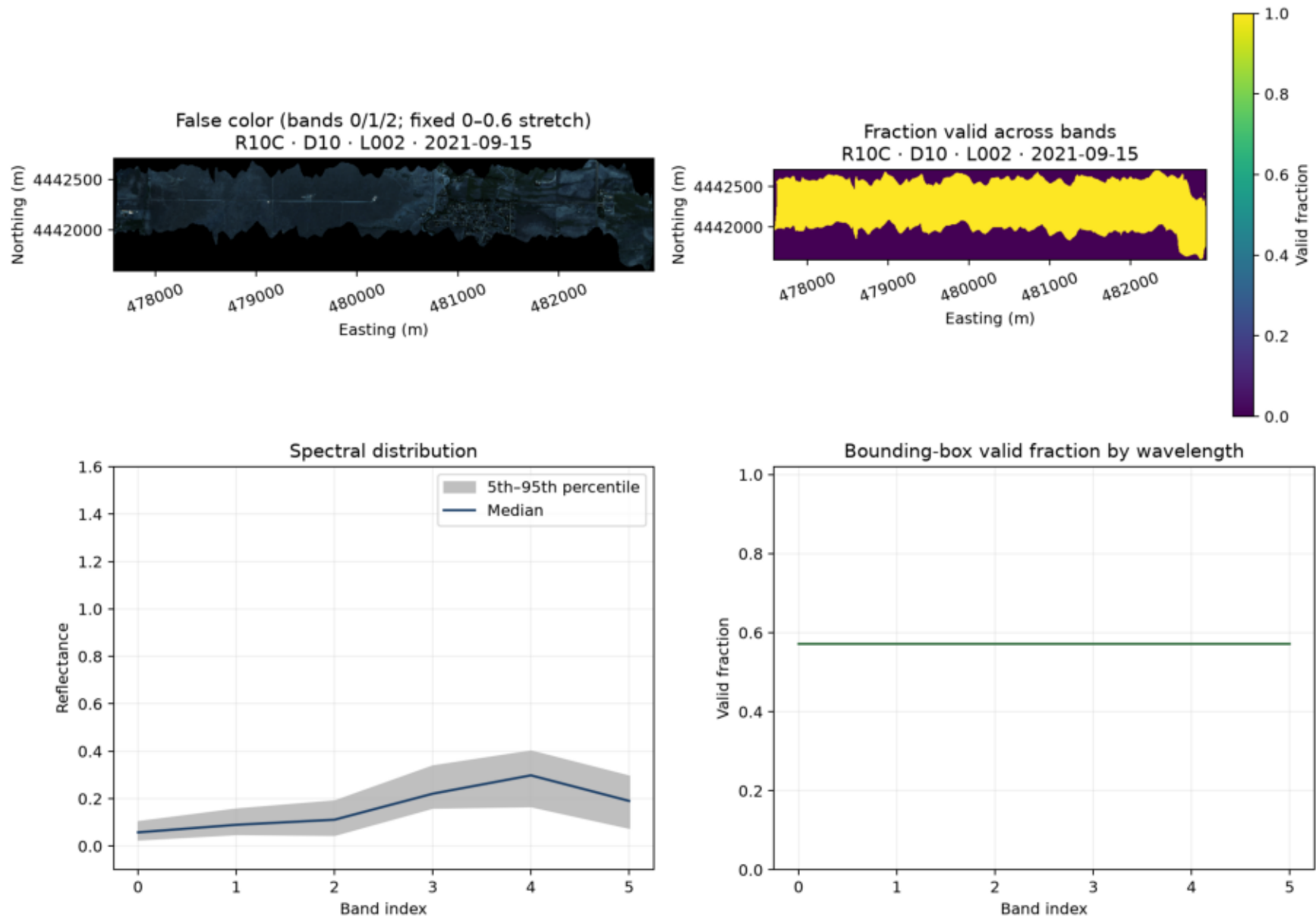
within_footprint_valid_reflectance_fraction: PASS | value=1 fraction | Valid spectral support should remain high inside the observed flight footprint; structural background outside it is reported separately. |

negative_reflectance_fraction: PASS | value=0 fraction | Large negative fractions may indicate correction or scaling

Spectral convolution Plot

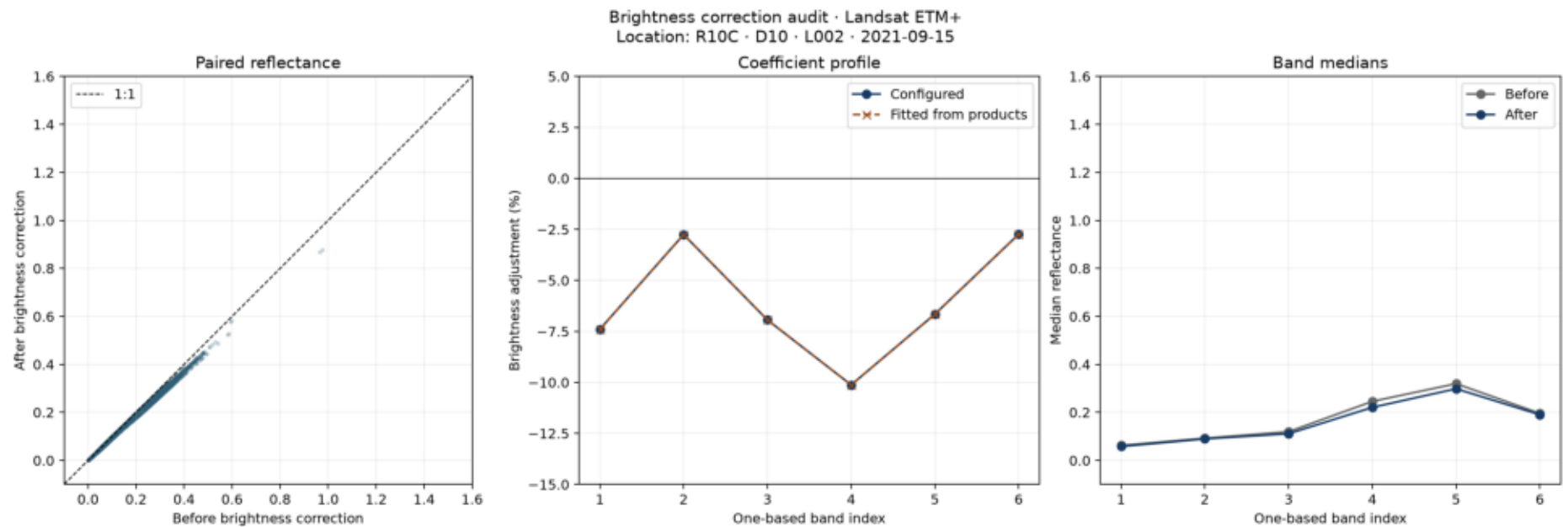
Spectral convolution: overview.png

Spectral convolution
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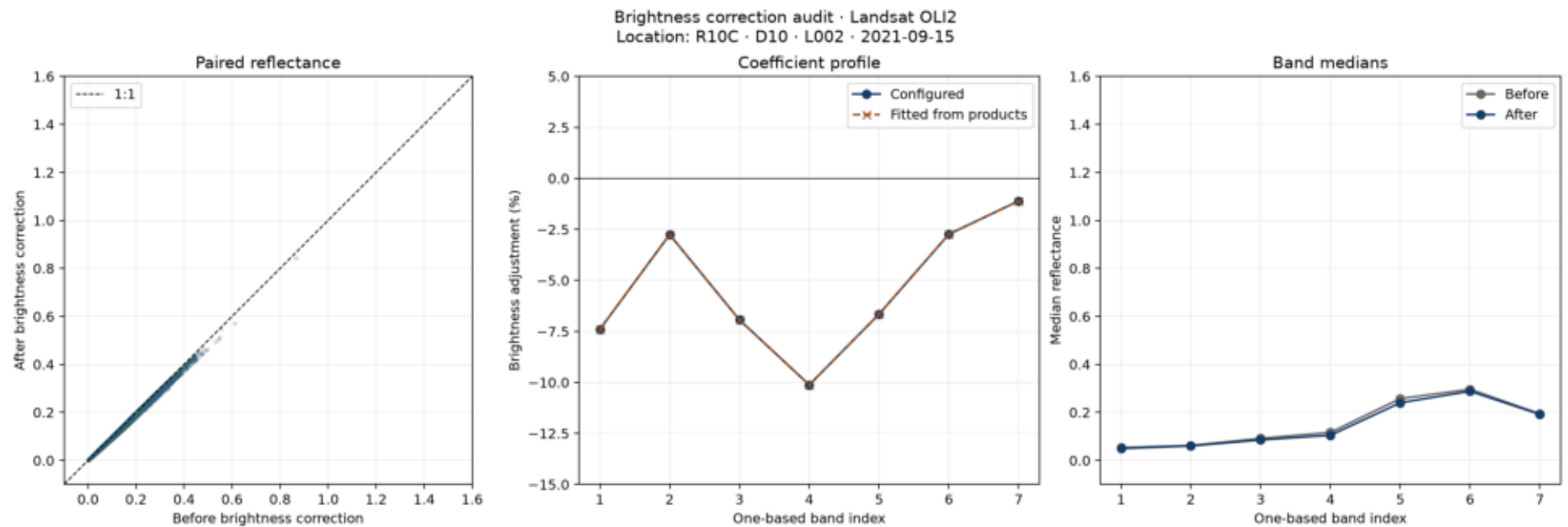
Spectral convolution Plot

Spectral convolution: brightness.png



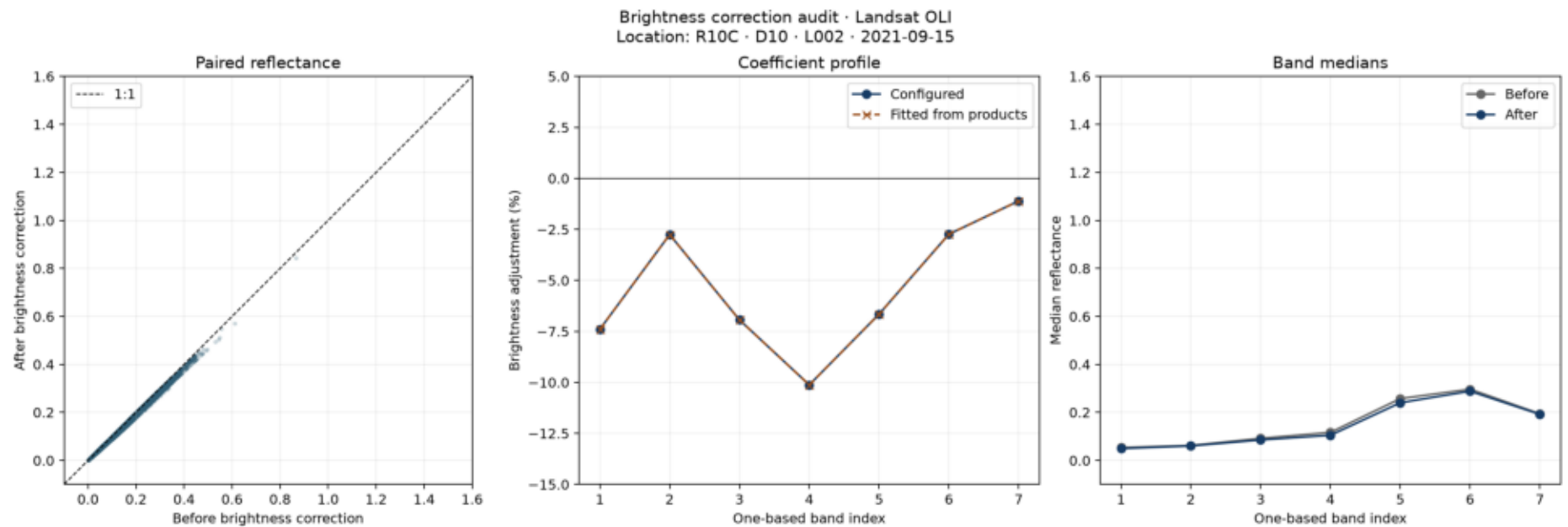
Spectral convolution Plot

Spectral convolution: brightness_02.png



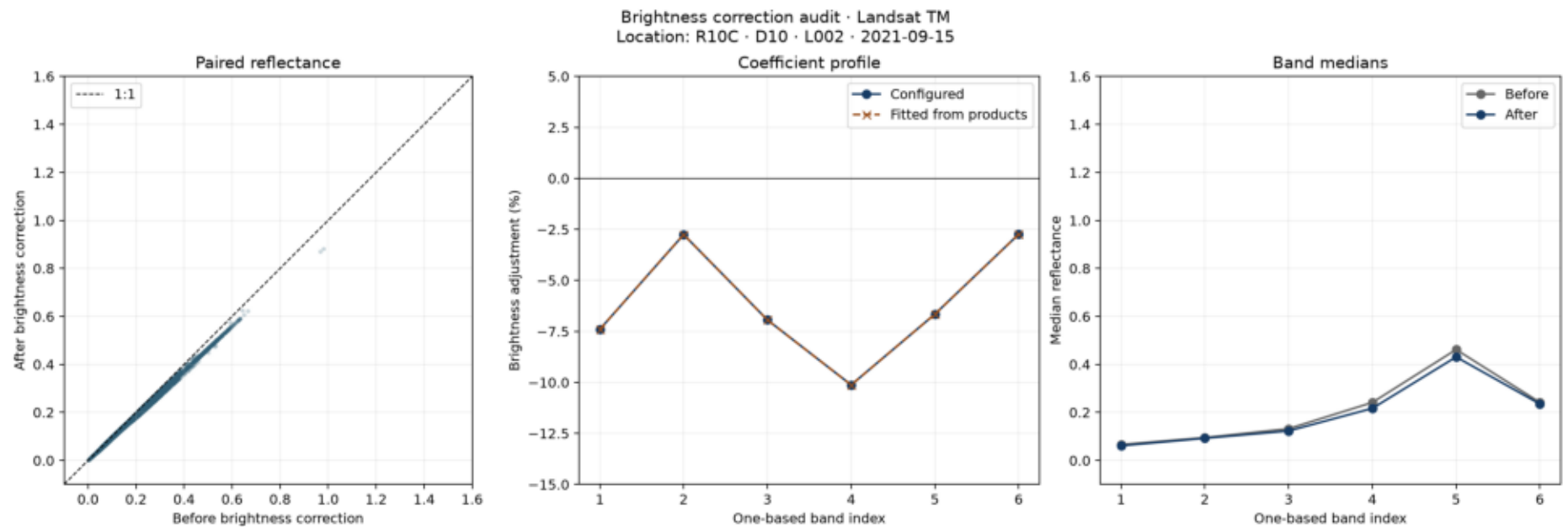
Spectral convolution Plot

Spectral convolution: brightness_03.png



Spectral convolution Plot

Spectral convolution: brightness_04.png



Parquet extraction and merge - PASS

Stage ID: analysis_tables | Mode: deep | Report: qa/stages/05_analysis_tables/stage_qa.html

Automated Interpretation

All evaluated checks passed their current thresholds.

Checks

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_brdfandtopo_corrected_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_etm+_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_etm+_undarkened_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli2_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli2_undarkened_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_oli_undarkened_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_tm_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_landsat_tm_undarkened_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_oli_oli2_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_oli_oli2_undarkened_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_tm_etm+_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

output_exists:NEON_D10_R10C_DP1_L002-1_20210915_directional_reflectance_micasense_to_match_tm_etm+_undarkened_envi_polygons.parquet: PASS | value=True | The canonical stage output exists on disk. |

Parquet extraction and merge Plot

Parquet extraction and merge: overview.png

